

Missing Value Handling – Explanation Table

Code 1

```
daily_data.isnull().sum()
```

Purpose

Checks whether the dataset contains any missing (NULL/NaN) values.

Function	Meaning	Use
<code>isnull()</code>	Identifies missing values	Detects empty cells
<code>sum()</code>	Counts missing values	Shows total missing values in each column

Example Output

Column	Missing Values
Sales_Amount	0
Units_Sold	0
Discount_Percent	5
Marketing_Spend	3
Customer_Footfall	0
Online_Orders	2
Economic_Index	1

Interpretation

Result	Meaning
0	No missing values
> 0	Missing values present

Viva Answer

The `isnull().sum()` function was used to identify and count missing values in each variable before model training.

Code 2

```
daily_data = daily_data.fillna(method='ffill')
```

Purpose

Replaces missing values using the previous valid observation.

Forward Fill (ffill)

Example:

Date	Sales_Amount
01-Jan	5000
02-Jan	NaN
03-Jan	5200

After Forward Fill:

Date	Sales_Amount
01-Jan	5000
02-Jan	5000
03-Jan	5200

Explanation Table

Function	Meaning	Use
<code>fillna()</code>	Fill missing values	Data cleaning
<code>method='ffill'</code>	Forward Fill	Copies previous valid value
<code>1'</code>		

Why Use Forward Fill?

Advantage	Reason
Preserves Data Size	No rows are deleted
Suitable for Time Series	Maintains continuity
Easy Implementation	Simple and effective
Prevents Model Errors	VARMAX cannot handle NaN values

Before and After

Stage	Missing Values
Before <code>fillna()</code>	Present
After <code>fillna()</code>	0

Verification:

```
daily_data.isnull().sum()
```

Expected Output:

Sales_Amount	0
Units_Sold	0
Discount_Percent	0
Marketing_Spend	0
Customer_Footfall	0
Online_Orders	0
Economic_Index	0

Project Report Table

Step	Code	Purpose
Missing Value Detection	<code>daily_data.isnull().sum()</code>	Identifies missing values in each column
Missing Value Treatment	<code>daily_data.fillna(method='ffill')</code>	Replaces missing values using previous observations
Verification	<code>daily_data.isnull().sum()</code>	Confirms dataset is free from missing values

Answer

Missing values were identified using `isnull().sum()` and handled using forward fill (`ffill`), which replaces missing observations with the previous valid value to maintain continuity in the time-series data before VARMAX modeling.

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